TMS INSIGHTS

Product Redesign Brief

A Unified Map-Centric Architecture for Collision Analytics

Document	TMS Insights — Product Redesign Brief	**Version**	1.0 — March 2026
Prepared for	City of Concord — Public Works / Traffic Safety Division	**Status**	Design Review Draft
Scope	Full platform audit, architecture redesign, persona-based build plan	**Personas**	5 personas defined

EXECUTIVE SUMMARY

TMS Insights is an AI-augmented application layer built on the City of Concord's Apache Superset collision analytics platform. This document identifies two structural data gaps, proposes a fundamental architecture shift that places the map as the primary interface, and defines a per-persona build plan across five user types. It covers the complete layer inventory of 27 geospatial data layers, four levels of map navigation (City, District, Corridor, Intersection), a redesigned Countermeasures workflow, a What-If Simulator with a grant document output pipeline, and an AI integration strategy built on structured component output rather than free-form text.

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01 CURRENT STATE ASSESSMENT

1. What TMS Insights Is Today

TMS Insights is an analysis layer on top of Apache Superset (Dashboard 26), extracting collision anomalies, amplifying risk signals, and presenting AI-generated interpretations to safety professionals. The underlying Superset platform holds the full collision dataset for the City of Concord, covering 4,777 incidents from 2014 to 2025, alongside 27 geospatial infrastructure and administrative layers.

The product currently operates as 15 independent analysis pages. Each page extracts a specific analytical view from the dataset. The core analytical depth is genuine and valuable. However, the product has significant structural problems that limit its usefulness and prevent it from serving three of the five user personas it needs to address.

1.1 The 15 Current Pages

Page	What it does	Disposition
Overview	City-level KSI summary + collage of excerpts from 5 other pages	**REPLACE — bottom half deleted, top becomes City panel**
Map Intelligence	Collision point map — severity, heatmap, cluster modes	**ABSORB into primary canvas**
Corridors	Corridor card wall — typology, KSI, trend per corridor	**ABSORB into Corridor level panel**
Collision DNA	10-archetype scatter plot of collision patterns	**PRESERVE as panel filter tool**
Risk Clock	Day x hour collision heatmap	**ABSORB into time slider**
Fatal Profile	Concord vs. County vs. State KSI benchmarks	**PRESERVE as City + District panel component**
Corridor Typology	Typology classification and comparison	**MERGE into Corridor Health Card**
What-If Simulator	CRF-based intervention modelling	**REDESIGN + PROMOTE to first-class workflow**
Countermeasures	Static countermeasure library with CRF values	**COMPLETELY REDESIGN**
Anomaly Rankings	Confidence vs. Impact scatter for anomalous locations	**ABSORB into map panel as view filter**
Amplification	Delta KSI by corridor + Pedestrian Action table	**MERGE into Corridor panel components**
Signatures	Corridor signature pattern classification	**MERGE into Corridor Health Card**
Temporal	Time-based collision trend analysis	**ABSORB into time slider + Risk Clock panel**
Silent Hotspots	Algorithmically flagged under-reported locations	**CONVERT to map layer toggle**
Conflict Matrix	Collision type x PCF cross-tab + Severity Model	**KEEP Severity Model only, in Factor Explorer**

1.2 What is Working

The analytical depth is real. The Factor Explorer, Collision DNA archetypes, Anomaly Rankings, Risk Clock heatmap, and the Severity Model in the Conflict Matrix are genuinely valuable analysis tools. The What-If Simulator runs correct CRF mathematics. The AI text blurbs, while passive, demonstrate that the right questions are being asked. None of this needs to be discarded — it needs to be repositioned within a map-centric architecture.

1.3 What is Not Working

- 15 separate pages create navigation overhead and force users to mentally assemble context that the product should maintain for them.
- The map is one of 15 pages, not the organizing principle. Every analysis in this product is ultimately about where something is happening — the map should be always present.
- The Overview bottom half duplicates 5 other pages without adding context. It reads as a forced collage, not a summary.
- AI is decorative. Text blurbs on each page describe what the charts already show. There is no AI that reasons across levels, responds to user context, or generates structured actionable output.
- Three personas — Engineers, Elected Officials, and the General Public — have no path through the product. The product serves only Safety Analysts and Corridor Directors.
- The Countermeasures page is a static reference library with no connection to specific locations, no ranking by relevance, and no output mechanism.
- The What-If Simulator produces a simulation but no document. Grant applications require a formatted submission package. The data and the math are there — the output is missing.

02 THE TWO CRITICAL GAPS

2. What TMS Insights Quietly Missed

Beyond the structural issues, two data dimensions are absent from TMS Insights that are essential to a complete collision safety product. These are not missing from the underlying data infrastructure — they are missing from the analytical layer that TMS Insights provides on top of Superset.

2.1 Gap One: Violations Data

Traffic citations — speeding, red light running, DUI stops, unsafe lane changes — are the pre-collision signal. They represent enforcement activity and observed driver behaviour that directly precedes collision outcomes. This data is absent from TMS Insights.

> *The PCF (Primary Collision Factor) codes already in the collision dataset describe what happened at the moment of a crash. Violations data captures the behavioural pattern that preceded it. These are different datasets telling different parts of the same story.*

Scenario	Collision Data Says	Violations Data Adds
High speed PCF, low crashes	Crashes are happening but infrequently	High citation density confirms problem; enforcement is partially working
High speed PCF, high crashes	Speed-related crash cluster identified	Low citation density = enforcement gap; structural speed problem not just behaviour
High DUI collisions on corridor	PCF shows alcohol as contributing factor	DUI citation mapping shows where enforcement is absent; targeted patrol argument
Traffic signal PCF violations	Signal violation is listed as PCF code	Red light citation data reveals whether it is signal timing or driver behaviour
Rising PCF but no crashes yet	No collision data available	Violations data shows early warning signal at an intersection before first crash

Violations data enables: a violation density map layer, violation type filtering aligned to PCF codes, enforcement gap analysis, pre-collision early warning, and — critically for the grant pipeline — a problem statement that combines crash history with enforcement evidence. Grant applications supported by both collision data and violation citation counts are materially stronger submissions.

2.2 Gap Two: Infrastructure Layers Are Invisible to the Analysis

The Apache Superset platform underlying TMS Insights has 27 fully populated geospatial layers across four groups: Collision, Community, Administrative Boundary, Facilities, and Assets. TMS Insights uses exactly one of them — the Collision point layer. The remaining 26 layers, representing the city's entire physical infrastructure inventory, never enter the analysis chain.

The connection that is missing — and that is central to every engineering and grant justification workflow — is: at locations where specific collision patterns occur, what infrastructure is present, what is absent, and what is proposed?

Collision Pattern	Infrastructure Gap Layer	Actionable Question Enabled
Pedestrian KSI cluster	Missing Sidewalk	Which pedestrian fatality locations have no sidewalk within 200ft?
Pedestrian KSI at night	Street Lights	Which dark-lighting KSI locations have no street light within 100ft?
Pedestrian right-of-way PCF	RRFB	Which pedestrian collision intersections have no RRFB or crosswalk signal?
Bicycle collisions on corridor	Bike Paths, Proposed Bike Routes	Where do bicycle crashes occur outside designated bike infrastructure?
School-hour collision peaks	Schools	Which school zones have collision clusters within 500ft of campus?
High PCF + no signal control	Traffic Signals, Stop Signs	Which uncontrolled intersections have the highest PCF collision counts?
Speed-related PCF	Proposed Speed Limits	Where are speed-related crashes occurring in zones with proposed limit reductions?

> *This is not a new dataset requirement. The data exists in Superset today. What is needed is proximity analysis logic that connects the collision point layer to the asset layers — and exposes the result through the TMS Insights interface.*

03 ARCHITECTURE RETHINK

3. The Map as the Central Canvas

The most important structural change in this redesign is moving the map from one page among fifteen to the organising principle of the entire product. Every analysis in TMS Insights is ultimately about where something is happening and what should be done about it. The map is the only interface that answers both questions simultaneously.

Charts, tables, and AI-generated text are supporting elements. They belong as panel content that responds to map state — not as independent pages that require the user to navigate away from the spatial context they were just building.

3.1 The New Structure

The new product has five functional views, down from fifteen pages:

View	What it contains	Replaces
Map (primary)	Full-screen map canvas with 4-level navigation, 27-layer system, right panel, AI, filters, time slider	Overview, Map Intelligence, Corridors, Risk Clock, Anomaly Rankings, Signatures, Temporal, Silent Hotspots, Amplification, Fatal Profile, Corridor Typology, Collision DNA
Countermeasures	Recommended, Applied, and Modelled countermeasures — location-aware, infrastructure-connected	Countermeasures (completely redesigned)
What-If Simulator	Three-panel simulation with Grant Package output	What-If Simulator (redesigned + output pipeline added)
Factor Explorer	Floating panel — 2-axis correlation + Severity Model + Violations dimension (when available)	Factor Explorer, Conflict Matrix (partial)
Grant Workspace	Active grant package drafts, export history, submission tracking	New — does not exist today

3.2 The Right Panel Principle

The map canvas always occupies the left 60% of the screen. The right 40% is a persistent context panel that updates as the map state changes. The panel is never blank — it always shows the most relevant analysis for the current level and entity. As the user drills from City to District to Corridor to Intersection, the panel deepens: each level adds a new layer of information while collapsing the previous level to a summary bar. This is the breadcrumb model — navigating back up restores the previous panel state without losing context.

> *For Safety Analysts jumping between corridors, this means they can return from an intersection back to a corridor view with all previously loaded context intact. Context persistence across levels is a critical usability requirement for experienced users who work quickly.*

04 FOUR-LEVEL NAVIGATION

4. The Navigation System: City to Intersection

The map operates at four levels. Level transitions are triggered by clicking or zooming — the same mental model as Google Maps. Each level has a distinct information state, filter context, and AI voice.

Level	Trigger	Right Panel Content	Default Layers Active	Primary Persona
City	Default entry / zoom out	Total KSI, YoY trend, top corridors, PCF distribution, benchmark vs. county/state	Collision (CSAP filter)	All personas
District	Click Voting District boundary or select from filter	District Scorecard: KSI rate, top 3 corridors, resident complaints, countermeasures deployed	Collision + Voting Districts + Resident Complaints	Elected Official
Corridor	Click corridor line or select from corridor list	Corridor Health Card: trend, PCF, typology, infrastructure status, Risk Clock, Violations (if available)	Collision + Corridor-relevant assets	Corridor Director, Engineer
Intersection	Click collision cluster or individual point	Intersection Profile: collision records, infrastructure within 100ft, complaints, countermeasures applied or modelled	All layers visible at this zoom	Safety Analyst, Engineer

4.1 Level Transitions

City to District: Clicking a Voting District boundary polygon zooms the map to that boundary and scopes all statistics within it. A district summary bar appears at the top of the right panel. The Voting Districts layer must be active — for the Elected Official persona preset, it is on by default.

District to Corridor: Clicking a corridor line within the district zooms to the corridor and opens the Corridor Health Card. The district summary bar collapses to a breadcrumb label at the top of the panel. Clicking the breadcrumb returns to the District level.

Corridor to Intersection: Clicking a collision cluster or individual point at street level opens the Intersection Profile. The corridor summary collapses to a breadcrumb. Clicking the breadcrumb returns to the Corridor level without reloading.

4.2 District Level: The Elected Official Entry Point

The Voting Districts layer is the Elected Official's primary navigation tool. When this layer is active, district boundaries appear as filled polygons on the map. Clicking a district boundary triggers the District Scorecard panel. The scorecard shows: KSI rate for this district vs. city average and county benchmark, year-over-year trend, top 3 colliding corridors by KSI count, number of resident complaints filed within the boundary, active countermeasures deployed in the past 2 years, and pending countermeasure requests. From the scorecard, the official can export a one-page district summary formatted for council presentation.

4.3 Corridor Level: The Corridor Health Card

The Corridor Health Card is the most used panel state for Corridor Directors and Engineers. It consolidates everything relevant to a single corridor: 3-year KSI trend (sparkline), dominant PCF category with count, corridor typology badge, a Risk Clock miniature heatmap for this corridor, infrastructure status summary (what assets are present and absent along the corridor), violation activity count if available, and resident complaints count. Two primary actions are available from this card: open Countermeasures (scoped to this corridor) and open What-If Simulator (pre-populated with this corridor's data).

4.4 Intersection Level: The Full Dossier

The Intersection Profile is the deepest level of the product. For each intersection or collision cluster, the panel shows: all collision records at this location (date, severity, PCF, MVIW, lighting, time), infrastructure inventory within a configurable proximity radius (default 100ft), any outstanding resident complaints tagged to this location, countermeasures already deployed (with date and before/after data if available), countermeasures currently modelled in the What-If Simulator, and the AI-generated intersection analysis. This profile is the starting point for a grant application. The "Begin Grant Package" button at the bottom of the profile opens the What-If Simulator pre-populated with this intersection's data.

05 THE COMPLETE LAYER SYSTEM

5. All 27 Layers — Organised by Persona Relevance

All 27 layers from the Superset platform are integrated into the TMS Insights map. They are organised in the layer panel into their natural groups. Each layer carries a persona relevance tag so that persona presets can activate the correct default stack.

Group	Layer	Role in Analysis	Relevant Personas
Base	**Collision**	Primary data layer. Always on. Severity colour-coding: red = fatal, orange = severe, yellow = visible, blue = complaint of pain.	All personas
Community	**Resident Complaints**	Community-reported safety concerns mapped by location. Connects public input to crash data.	General Public, Elected Official
Admin	**Voting Districts**	Electoral boundary polygons. The Elected Official's primary navigation layer and the District Scorecard trigger.	Elected Official
Admin	**Zoning District**	Land use context for collision location — residential, commercial, industrial.	Engineer, Elected Official
Admin	**General Plan**	Long-range planning designations. Connects collision patterns to future development context.	Engineer, Elected Official
Admin	**Parks**	Park boundaries. Pedestrian generator — relevant for pedestrian collision proximity analysis.	General Public, Elected Official
Admin	**Cal Enviro Property**	Environmental justice screening areas. Equity dimension for grant applications and elected official reporting.	Elected Official, Engineer
Facilities	**Schools**	School locations. Safe Routes to School analysis. High pedestrian generator during school hours.	General Public, Elected Official, Engineer
Facilities	**Hospitals**	Hospital proximity for severity response context. Relevant for pedestrian and cyclist collision clusters.	Safety Analyst, Engineer
Facilities	**Shopping Centers**	High pedestrian and vehicle generator. Relevant to parking lot and access road collision analysis.	General Public
Facilities	**City Facilities**	Government-owned sites. Relevant for jurisdiction context and countermeasure approval pathways.	Corridor Director, Elected Official

Facilities	**BART Track**	Transit rail alignment. Relevant for at-grade crossing collision analysis.	Engineer, General Public
Facilities	**BART Station**	Transit access nodes. High pedestrian generator. Relevant for pedestrian and cyclist collision proximity.	Engineer, General Public
Assets	**Bus Stops**	Transit stop locations. Pedestrian generator. Relevant for pedestrian collision proximity and lighting analysis.	Engineer, Safety Analyst
Assets	**Bike Paths**	Existing designated bicycle infrastructure. Relevant for cyclist collision gap analysis.	Engineer, Safety Analyst
Assets	**Proposed Bike Routes**	Planned bicycle infrastructure. Shows where the city intends to invest — relevant to current collision patterns.	Engineer, Corridor Director
Assets	**Proposed Speed Limits**	Planned speed limit changes. Connects speed-reduction proposals to existing speed-related collision clusters.	Engineer, Corridor Director
Assets	**Missing Sidewalk**	Locations where sidewalk is absent. PRIMARY engineering gap layer for pedestrian collision analysis.	Engineer, Safety Analyst
Assets	**Handicap Ramps**	ADA-compliant ramp locations. Relevant to pedestrian mobility and crosswalk safety analysis.	Engineer
Assets	**Bike Parking**	Bicycle parking locations. Relevant to cyclist trip generation and collision context.	Engineer
Assets	**RRFB**	Rectangular Rapid Flashing Beacons. Key pedestrian protection device. Gap analysis: where are RRFBs absent at pedestrian KSI locations?	Engineer, Corridor Director, Safety Analyst
Assets	**Sign Flashers**	Flashing warning sign locations. Relevant to intersection safety and speed advisory.	Engineer, Corridor Director
Assets	**Stop Signs**	Stop-controlled intersections. Relevant to right-of-way PCF collision analysis.	Engineer
Assets	**Traffic Signals**	Signalised intersections. Critical for signal violation PCF analysis and What-If simulation.	Engineer, Corridor Director
Assets	**Street Lights**	Street lighting locations. Critical for dark-condition KSI analysis. Gap: KSI at dark lighting with no street light within 100ft.	Engineer, Safety Analyst

Assets	**Vehicle Detection Systems**	Intersection detection hardware. Relevant to signal timing analysis and conflict matrix.	Safety Analyst, Corridor Director
Assets	**Parking Lots**	Off-street parking areas. Relevant to access road conflict and pedestrian path of travel.	General Public

5.1 Persona Presets

When a user selects their persona at entry, the layer panel activates the recommended default stack for that role. All other layers remain available but are off by default. Users can always override.

Persona	Default Active Layers
Safety Analyst	Collision + Vehicle Detection Systems + RRFB + Traffic Signals + Street Lights + Missing Sidewalk
Corridor Director	Collision + Proposed Speed Limits + Sign Flashers + RRFB + Traffic Signals + Vehicle Detection Systems
Engineer	Collision + Missing Sidewalk + Proposed Bike Routes + Proposed Speed Limits + RRFB + Street Lights + Traffic Signals + Handicap Ramps
Elected Official	Collision + Voting Districts + Schools + Parks + Resident Complaints + Cal Enviro Property
General Public	Collision (simplified severity labels) + Schools + Parks + Shopping Centers + BART Station + Resident Complaints

06 PERSONA-BASED EXPERIENCE DESIGN

6. Five Personas — Five Distinct Experiences

The product redesign serves five distinct user personas. Each has a different entry point, a different default map state, different primary workflows, and a different primary output. The underlying data and analysis engine are shared — only the surface experience differs.

6.1 Safety Analyst

Entry state: City level, CSAP filter active (Fatal + Severe + Visible Injury, 2021+). Full layer access. Heatmap overlay + anomaly markers as default.

Primary workflows:

- Anomaly investigation — map shows flagged intersections as pulsing anomaly markers. Click to drill to Intersection level. AI panel shows anomaly score, contributing factors, z-score context, and infrastructure gap if applicable.
- Factor correlation — select two factors in the Factor Explorer (PCF x MVIW, Lighting x Collision Type) to generate a correlation heatmap that colours the map itself, not just a separate chart.
- Temporal sweep — Risk Clock time slider animates the map through 24 hours. Severity density changes in real time.
- Violations overlay — toggle violation density layer to identify enforcement gaps versus collision clusters when violations data is available.

Primary output:

- Anomaly flags with infrastructure context attached. AI-generated factor summary. Watchlist export for handoff to Corridor Director or Engineer.

6.2 Corridor Director / Corridor Safety Analyst

Entry state: City level, then select a corridor. Jumps directly to Corridor level (Level 3).

Primary workflows:

- Corridor review — the Corridor Health Card shows everything for the assigned corridor in a single panel: KSI trend sparkline, PCF breakdown, typology badge, Risk Clock miniature, infrastructure status, complaints count.
- Countermeasure selection — from the Corridor Health Card, one click opens the Countermeasures panel scoped to this corridor. Recommended interventions are pre-ranked by projected KSI impact for this specific corridor.
- What-If simulation — pre-populated simulator from the Corridor Health Card. Run simulations and review projected KSI reduction.
- Field briefing — save Corridor Health Card snapshot as a one-page PDF for field team or internal review.

Primary output:

- Corridor action plan. What-If simulation result. Field briefing one-pager.

6.3 Traffic / Civil Engineer

Entry state: Engineer persona preset auto-activates. City level with Missing Sidewalk + RRFB + Traffic Signals + Street Lights visible. This persona is completely new to the product.

Primary workflows:

- Infrastructure Gap Analysis — dedicated view within the map panel. Ranked list of locations where a specific infrastructure type is absent and collision data justifies intervention. "Show all pedestrian KSI locations with no RRFB within 200ft." The result is a ranked work queue exportable as a table.
- Countermeasure scoping — at each flagged location, select applicable countermeasures from the Recommended panel. Build a project package for one or multiple locations.
- What-If Simulation to Grant Package — run the simulation, generate the full grant request document. This is the primary output of the Engineer workflow.

Primary output:

- Infrastructure gap report. Grant Request Package document. HSIP/ATP submission ready.

6.4 Elected Official (District)

Entry state: Voting District selection screen. Once selected, all data scopes to the district boundary. This persona is entirely new to the product.

Primary workflows:

- District Scorecard — KSI rate vs. county and state benchmarks. Year-over-year trend. Top 3 corridors. Resident complaints count. CalEnviro equity overlay for vulnerable population context.
- Infrastructure lens — Schools + BART layers show pedestrian vulnerability concentrations. Cal Enviro Property shows equity context.
- What's Being Done — countermeasures deployed in the district vs. requested vs. pending. Shows investment and progress, not just risk.
- Constituent summary — one-page district report exportable for council presentations or constituent communications.

Primary output:

- District summary report. Council presentation slide content. Constituent communication.

6.5 General Public

Entry state: Address search or "use my location." Map zooms to street level. Simplified interface — no filter bar visible by default. No jargon, no KSI codes, no z-scores.

Primary workflows:

- My street — collision history within a 500ft radius shown as plain-language severity labels ("1 fatal crash," "3 injury crashes" — not "1 KSI Fatal, 3 KSI Non-Fatal").
- What's here — infrastructure visible on the map (schools, BART, parks, shopping centres).
- Resident complaints — view existing complaints near the location. File a new complaint with a structured form. Complaint appears on the map immediately.
- What's being done — deployed countermeasures shown nearby with plain language description ("A pedestrian signal was installed at this intersection in 2024").

Primary output:

- Resident complaint submission. Increased community awareness.

07 COUNTERMEASURES — COMPLETE REDESIGN

7. Countermeasures: From Library to Recommendation Engine

The current Countermeasures page is a static reference: a list of interventions with CRF values. It has no connection to specific locations, no ranking by relevance, no before/after evidence from local deployments, and no output path. The redesigned screen operates in three states and is always scoped to the currently selected map entity.

7.1 State One: Recommended

When a corridor or intersection is selected on the map, the Countermeasures panel automatically generates a ranked list of applicable interventions. Ranking logic:

- Collision type match: does this countermeasure address the dominant collision type at this location?
- PCF match: does this countermeasure target the observed primary collision factors?
- Infrastructure gap match: does this countermeasure address an identified infrastructure gap at this location?
- CRF value: projected KSI reduction percentage from authoritative source.
- Local evidence: has this countermeasure been deployed elsewhere in Concord, and what did the before/after data show?

> *Each Recommended card explicitly states the infrastructure connection: "3 pedestrian KSI collisions at this location in 3 years. No RRFB within 300ft. Nearest RRFB is 0.4 miles east on Concord Blvd. RRFB CRF for pedestrian collisions: 0.47 (47% KSI reduction). 2 similar RRFB installations in Concord showed 38% and 51% reduction over 2-year post-periods."*

7.2 State Two: Applied

All countermeasures already deployed within Concord, with location, installation date, and before/after collision data where available. This view serves two purposes: it gives Engineers and Corridor Directors local evidence of what works (not just national CRF averages), and it gives Elected Officials a record of the city's investment and impact. Each Applied card links to the map location so the user can see it in spatial context.

7.3 State Three: Modelled

Countermeasures currently staged in the What-If Simulator. Shows the full portfolio of modelled interventions across all locations, their individual projected outcomes, and their combined city-wide KSI reduction estimate. A direct "Generate Grant Package" button connects to the Grant Workspace for any modelled intervention. This state is the bridge between the simulation and the submission document.

7.4 Card Anatomy

Card Element	Content
Countermeasure Name	Plain language name + formal engineering designation
CRF Value	Percentage KSI reduction from authoritative source (FHWA CMF Clearinghouse)
Target Collision Types	Which collision types and PCF codes this intervention addresses

Infrastructure Gap Addressed	Which missing asset layer this countermeasure would fill or supplement
Local Evidence	Deployments in Concord with before/after data. If none, shows comparable city examples.
Cost Estimate	Range from comparable Concord or regional deployments
Simulate Button	Opens What-If Simulator pre-populated with this countermeasure and the current location's data

08 WHAT-IF SIMULATOR & GRANT PIPELINE

8. The What-If Simulator: The Grant Pipeline Golden Leg

The What-If Simulator is the highest-value feature in the product for Engineers and Corridor Directors. Its current implementation runs the simulation correctly but produces no actionable output document. The redesigned simulator has three panels and a direct document generation pipeline.

> *The Superset API response already carries a grant_exportable flag on each data record. The data infrastructure for grant document generation is in place. What is needed is the document assembly and formatting layer.*

8.1 Simulator Interface — Three Panel Layout

Left panel — Location and Baseline: The selected corridor or intersection, current 3-year collision stats, severity breakdown, top 3 PCF categories, and infrastructure status summary. Pre-populates automatically from the map selection. This panel is read-only — it is the evidence base.

Centre panel — Countermeasure Configuration: One or multiple countermeasures selected from the library. Each shows its CRF applied specifically to the collision types observed at this location (not a generic CRF — a location-adjusted projection). As countermeasures are added or removed, the simulation recalculates in real time. Multiple countermeasures show individual and combined projected outcomes.

Right panel — Projected Outcome: KSI count before and after. Percentage reduction. Fatalities prevented over a 5-year projection window. Injuries prevented. Equity of benefit — which severity levels and collision types benefit most. A confidence indicator showing the statistical basis for the projection. A note on data limitations where relevant.

8.2 The Grant Package — Document Contents

From any simulation state, "Generate Grant Package" assembles a formatted document containing the following sections:

Section	Content Source
1. Project Title and Location	Auto-generated from corridor/intersection name + countermeasure type
2. Problem Statement	Current collision data at the location, benchmarked against city average and county rate. Plain language framing. Violations data included if available.
3. Proposed Intervention	Selected countermeasure(s) with description, CRF sourcing citation, and infrastructure gap context
4. Expected Outcome	KSI reduction projection from What-If Simulator. 5-year fatalities prevented. 5-year injuries prevented. Cost-benefit estimate.
5. Supporting Data Tables	Collision data table by year (3-year lookback). Severity breakdown. Lighting conditions. PCF breakdown. Time-of-day analysis (Risk Clock data for this location).
6. Map Exhibit	Exported map view of the location with collision points, severity colour-coding, and relevant infrastructure layers visible
7. Equity Statement	Cal Enviro Property designation if applicable. Affected populations identified (schools, BART stations, pedestrian routes within 500ft proximity).
8. Cost Estimate	Range pulled from comparable Applied deployments in Concord or regional database

9. Grant Program Alignment	Alignment statement for HSIP, ATP, or Caltrans Highway Safety grant requirements as selected by the user

8.3 Grant Programs Supported

- HSIP — Highway Safety Improvement Program (Federal): Requires crash history, countermeasure CRF, projected fatality reduction, and cost-effectiveness ratio.
- ATP — Active Transportation Program (California): Requires pedestrian and cyclist collision data, infrastructure gap documentation, equity analysis (Cal Enviro), and community support evidence.
- Caltrans Highway Safety: Requires location-specific crash data, engineering justification, and projected outcome table.
- Resident Complaints data and Cal Enviro Property data directly strengthen ATP applications by providing community need evidence and equity documentation — two layers already in the system.

09 AI INTEGRATION STRATEGY

9. AI as a Product Layer, Not a Text Blurb

AI in the current product is a static text blurb on each page — it describes what the charts already show. In the redesigned product, AI operates as a dynamic, context-aware layer across the entire map canvas. Its outputs are not free-form text — they are structured components that render predictably in the UI.

9.1 Three Interaction Modes

Ambient (Passive)

The AI panel on the right side of the map auto-populates whenever the map state changes: new level selected, new entity loaded, new filter applied, time slider moved. The content matches the level — city-level summary at city view, corridor health narrative at corridor level, intersection detail at intersection level. The user never has to ask for it.

Select and Invoke

Any data element visible on the map or in the panel can be selected to trigger a contextual AI action ring with four options: Explain (plain language interpretation of the selected element), Compare (run comparison to similar corridors, locations, or typology peers), Simulate (open What-If Simulator pre-populated), Flag (add to watchlist with AI-generated reason). This is the "select a thing, ask about that thing" model that replaces generic chat for expert users.

Floating Contextual Chat

A persistent floating chat panel, aware of the current map state at all times. Context payload sent with every message: current level, selected entity, active time window, visible layers, active filters. Users ask questions in natural language. The chat can answer questions that span multiple levels ("Which intersections in this district have no RRFB and pedestrian KSI?"), drive map state changes ("Show me dark-condition fatals on Clayton Road"), and connect to the simulator ("What would adding RRFB at the top 5 pedestrian locations cost?").

9.2 AI Output Format — The Component Library

AI responses are constrained to a structured component library. The AI returns a JSON payload specifying which components to render. This prevents text blobs and ensures every AI output is consistently rendered, visually coherent, and actionable. The 12 defined component types:

Component	Renders As	Example Use
stat_chip	Number + label + trend arrow	KSI count this year vs. prior year
comparison_bar	Two-value horizontal bar	This corridor vs. city average KSI rate
anomaly_card	Flagged location card with score	Top anomalous intersection with z-score
trend_spark	Mini sparkline chart	Monthly collision trend for selected corridor
risk_band	Time-of-day risk bar	Peak crash hours for this location
factor_tag	Coloured label chip	Primary PCF at this intersection
countermeasure_chip	Recommendation card with CRF	RRFB recommended — 47% KSI reduction

benchmark_flag	Comparison badge	Clayton Rd is 2.3x the city average
narrative_sentence	Plain text insight	Contextual explanation for non-technical users
radar_mini	Mini radar chart	Corridor risk profile across 6 dimensions
collision_list	Recent incidents table	Last 5 collisions at this intersection
typology_badge	Corridor type label	High-Speed Commercial Corridor

> *The AI never outputs a paragraph of unstructured text. It returns {level, entity, components: [{type, ...props}]}. The frontend renders the components. This is the design contract between the AI system and the product designer.*

010 BUILD PRIORITIES

10. What to Build — Phased Plan

The redesign is structured in three phases. Phase 1 establishes the map-centric architecture and makes the product usable for all five personas. Phase 2 activates the Countermeasures and Grant pipeline — the two highest-value new capabilities. Phase 3 adds violations data and expands the public-facing experience.

Phase 1 — Foundation

Objective: Map-centric architecture, all 5 personas served, infrastructure correlation active.

1. Replace the 15-page structure with the unified map canvas. The map always occupies 60% of the screen. The right panel always reflects the current map state.
2. Implement four-level navigation (City, District, Corridor, Intersection) with breadcrumb context persistence.
3. Integrate all 27 Superset layers into the TMS Insights map with the layer panel organised by group.
4. Build persona preset system — five role configurations that activate the correct layer stack and default filter state on entry.
5. Build the Corridor Health Card (Level 3 panel) consolidating trend, PCF, typology, infrastructure status, and Risk Clock.
6. Build the Intersection Profile (Level 4 panel) with collision records, infrastructure inventory, and complaints.
7. Implement proximity analysis engine: for any selected location, identify which asset types are present or absent within a configurable radius.
8. Implement Voting Districts boundary layer and District Scorecard panel for Elected Official entry.
9. Implement Resident Complaints layer (Community data from Superset) with complaint submission form for General Public.
10. Implement ambient AI panel — context payload (level, entity, timeWindow, visibleLayers) sent on every map state change. AI returns component JSON.

Phase 2 — Countermeasures and Grant Pipeline

Objective: Countermeasures become a recommendation engine. What-If Simulator produces a grant document.

11. Redesign Countermeasures screen with three states: Recommended (location-aware, ranked), Applied (local evidence), Modelled (simulator staging).
12. Build countermeasure recommendation engine: collision type + PCF match + infrastructure gap match + CRF ranking.
13. Redesign What-If Simulator as three-panel layout (Location/Baseline | Countermeasure Config | Projected Outcome).
14. Build Grant Package document generator producing a formatted, submission-ready document from any simulation state.
15. Build Grant Workspace view — active packages, export history, grant program alignment selector.
16. Implement Select-and-Invoke AI action ring (Explain, Compare, Simulate, Flag) on all map and panel elements.
17. Implement floating contextual chat with map-state-aware context payload.

Phase 3 — Violations Data and Extended Personas

Objective: Add the missing pre-collision signal. Full public and elected official experiences.

18. Integrate violations data from CPD data pipeline or SWITRS (requires data agreement and ETL pipeline).
19. Add Violations as a map layer (density view by violation type) and as a Factor Explorer dimension.
20. Build Engineer persona Infrastructure Gap Analysis view — ranked list query interface ("Show locations where X infrastructure is absent and Y collision type exceeds Z threshold").
21. Build General Public mobile-optimised map view — simplified labels, address search, complaint submission, plain-language AI narrative.
22. Build Elected Official one-page district report export (PDF, formatted for council presentation).
23. Extend Grant Package with violations data section where available — problem statement, enforcement gap narrative, combined citation + crash evidence.

011 DESIGN PRINCIPLES

11. What to Think Differently

Beyond the specific feature changes, these are the underlying mental model shifts that should guide every design decision in this product.

The map is not a feature — it is the product

Every page in the current product is a chart on top of data. The redesigned product is a map with analysis attached. When a designer makes a decision about where something lives, the first question is: does this belong on the map canvas, in the right panel, or in a dedicated workflow (Countermeasures, Simulator, Grant Workspace)? If it can be expressed on the map, it should be.

Infrastructure absence is as important as collision presence

The current product shows where crashes happen. The redesigned product shows where crashes happen and what is missing at those locations. These are different analytical statements. The missing sidewalk, the absent RRFB, the dark street with no light — these are the engineering story that the collision data alone cannot tell. Every screen that shows a collision cluster should also be able to show the infrastructure context of that cluster.

AI produces components, not commentary

The text blurb model of the current product treats AI as a narrator. The redesigned product treats AI as a renderer. AI takes context and produces structured output that the UI assembles into coherent panels. This is not an aesthetic preference — it is a reliability and consistency requirement. When AI can only output from a defined component library, every response is predictable, reviewable, and actionable. Free-form AI text in a safety-critical product is a liability.

The grant letter is the product's highest-value output

Every Engineer and Corridor Director who uses this product is ultimately trying to get funding to fix something. The What-If Simulator's CRF calculations and the collision data are the inputs to a grant application. The product should produce that application. Not as a nice-to-have feature — as the primary output of the engineering workflow. If a grant submission can be drafted in the product in under 30 minutes using real data, that is transformative.

Context persistence is respect for the expert user

Safety analysts and corridor directors are experts. They work fast, they hold multiple corridors in mind simultaneously, and they hate losing their place. The breadcrumb navigation model, the persistent right panel, and the context-aware AI chat are not UX enhancements — they are the baseline expectation for a professional-grade tool. Every navigation action should either deepen context or explicitly return to a prior state. Nothing should clear context unexpectedly.

The general public view is not a simpler version of the analyst view

It is a different product with different goals, built on the same data. The General Public experience should never show a z-score, a KSI rate, or a PCF code. It should show: crashes near you, infrastructure near you, what people have reported, and what is being done. The Resident Complaints layer is the connective tissue between the public and the analytical product — when a resident's complaint appears on the same map as a crash cluster and a pending countermeasure request, the city becomes legible.

END OF DOCUMENT

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